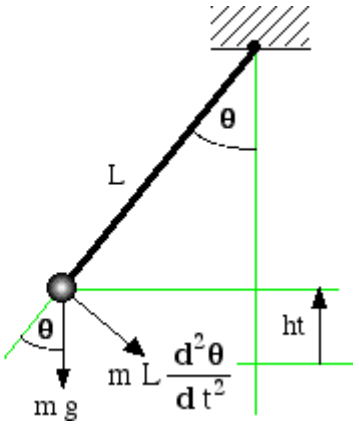
Sec 8.6 Representing functions with power series.

Some times it is useful to approximate a function with a power series. One example of this is the differential equation for a pendulum.

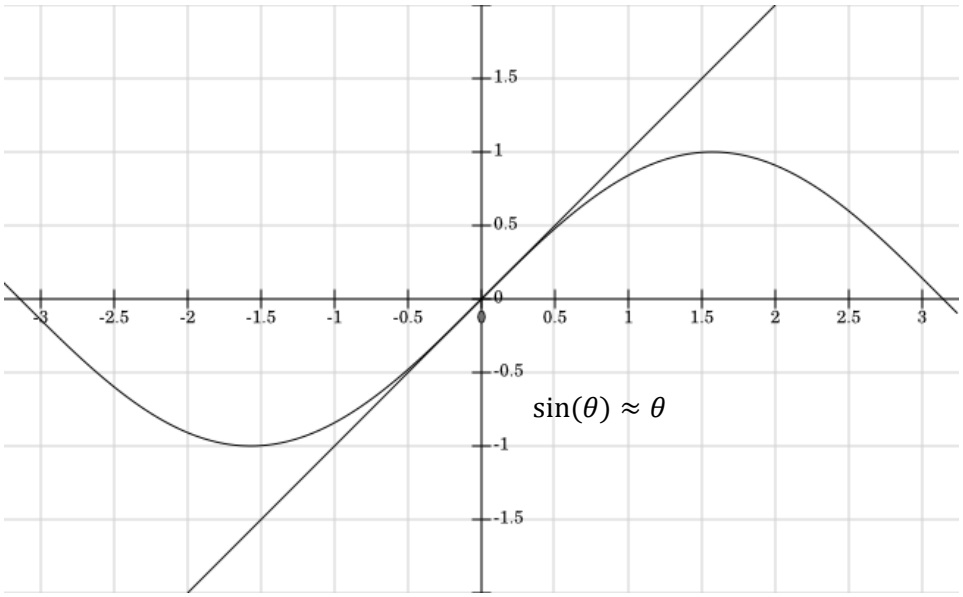
From a force balance we obtain:

$$m\,g\,\sin(\theta) + m\,L\,\frac{d^2\theta}{dt^2} = 0$$

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\sin(\theta)$$



It would be much easier to solve this if $\sin(\theta)$ could be approximated by just θ. In fact we can! (At least for small angles.)



$$\frac{d^2\theta}{dt^2} = -\frac{g}{l}\sin(\theta)$$

In fact

$$sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!} + \cdots$$

http://fooplot.com/#W3sidHlwZSI6MCwiZXEiOiJzaW4oeCkiLCJjb2xvciI6IiMwMDAwMDAifSx7InR5cGUiOjAsImVxIjoieCIsImNvbG9yIjoiIzAwMDAwMCJ9LHsidHlwZSI6MCwiZXEiOiIiLCJjb2xvciI6IiMwMDAwMDAifSx7InR5cGUiOjEwMDAsIndpbmRvdyI6WyItMy4yNSIsIjMuMjUiLCItIMiLsIjIiXX1d

Sec 8.6 Representing functions with power series.

In Fact any continuous differentiable function can be represented by a power series! There are many ways to obtain them one of which is to start with our old friend the geometric series. When |x|<1 we have that

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + x^5 + \cdots$$

Let's use this series to find a power series for the function $f(x) = \frac{1}{1+x^2}$

So as long as |x|<1

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + x^8 - x^{10} + \cdots = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}$$

Sec 8.6 Representing functions with power series.

We can also Integrate or differentiate series to obtain other series ...

Ex Starting with the Power series for the geometric series find a power series for $\tan^{-1}(x)$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + x^5 + \cdots = \sum_{n=0}^{\infty} x^n$$

Sec 8.6 Representing functions with power series.

Starting with the power sereis for the geometric series find a power series for the function $f(x) = \frac{1}{(1-x)^2}$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + x^5 + \cdots = \sum_{n=0}^{\infty} x^n$$

Sec 8.6 Representing functions with power series.

2 Theorem If the power series $\Sigma\, c_n(x-a)^n$ has radius of convergence $R > 0$, then the function f defined by

$$f(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \quad = \sum_{n=0}^{\infty} c_n(x-a)^n$$

is differentiable (and therefore continuous) on the interval $(a-R, a+R)$ and

(i) $f'(x) = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + \quad = \sum_{n=1}^{\infty} nc_n(x-a)^{n-1}$

(ii) $$\int f(x)\,dx = C + c_0(x-a) + c_1\frac{(x-a)^2}{2} + c_2\frac{(x-a)^3}{3} +$$

$$= C + \sum_{n=0}^{\infty} c_n\frac{(x-a)^{n+1}}{n+1}$$

The radii of convergence of the power series in Equations (i) and (ii) are both R.

This can help tremendously in dealing with tough integrals!

Ex Integrate $\int \frac{1}{1-x^5}\,dx$

Sec 8.6 Representing functions with power series.

HW Example Sec 8.6 #25

Use a power series to evaluate the integral. What is the radius of convergence?

$$\int \frac{x - tan^{-1}(x)}{x^3} dx$$